

FPGA Accelerator for Directed Percolation

Monte Carlo Simulation on Artix-7 at 100 MHz

Leonardo Pieripolli, Alessio Tuscano

Programmable Hardware Devices

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Outline

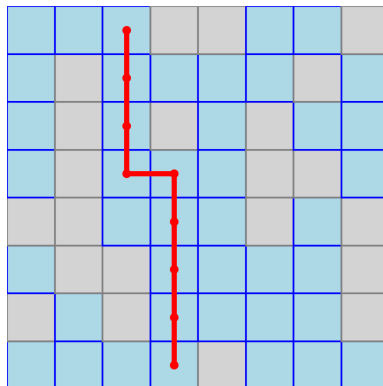
- 1 Percolation Theory
- 2 System Architecture
- 3 Validation
- 4 FPGA Engineering Results
- 5 Physics Results
- 6 Conclusions
- 7 Backup

What is Percolation?

Physical picture: a fluid filters through a porous medium.

- 2D grid: each site is **open** (blue, prob. p) or **closed** (gray, $1 - p$)
- A **spanning cluster** connects top $\rightarrow$ bottom
- **Phase transition** at critical probability p_c
- Below p_c : only finite clusters
Above p_c : spanning cluster emerges
- **Isotropic percolation:** fluid can move in all 4 directions

Isotropic Percolation



Directed Percolation (DP)

Key difference: flow is **directional** — only downward propagation.

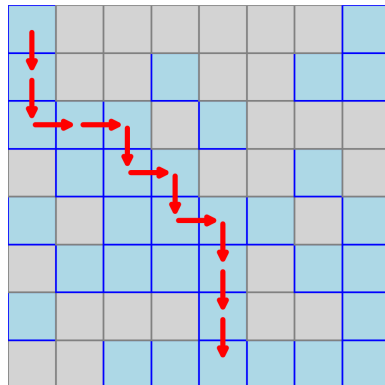
Directed Percolation

- Fluid moves only $\downarrow$ (not $\uparrow$)
- Horizontal spread still allowed ($\leftrightarrow$)
- Different universality class from isotropic percolation
- DP critical exponents: useful to find cluster mass and p_c for example

$$\beta = 0.2765 \quad (\text{order parameter})$$

$$\nu_{\parallel} = 1.096 \quad (\text{corr. length, time})$$

$$\nu_{\perp} = 0.733 \quad (\text{corr. length, space})$$



Finite-Size Scaling

For a finite grid of size $N \times N$, the spanning probability obeys:

$$P_{\text{span}}(p, N) = F\left((p - p_c) N^{1/\nu}\right)$$

where F is a **universal scaling function**.

Consequences:

- Curves for different N **collapse** onto a single master curve
- The collapse validates the universality class
- The crossing point of $P_{\text{span}} = 0.5$ gives an estimate of p_c

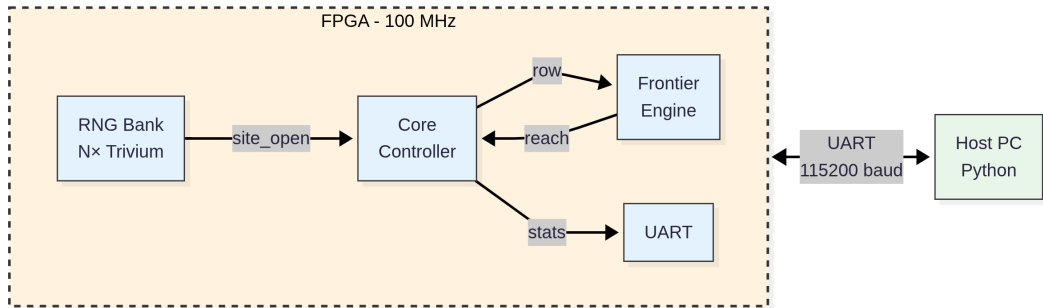
Threshold estimation via logistic regression:

$$P_{\text{span}}(p) = \frac{1}{1 + e^{-a(p-p_c)}}$$

Fitted via maximum likelihood (binomial bootstrap for CIs).

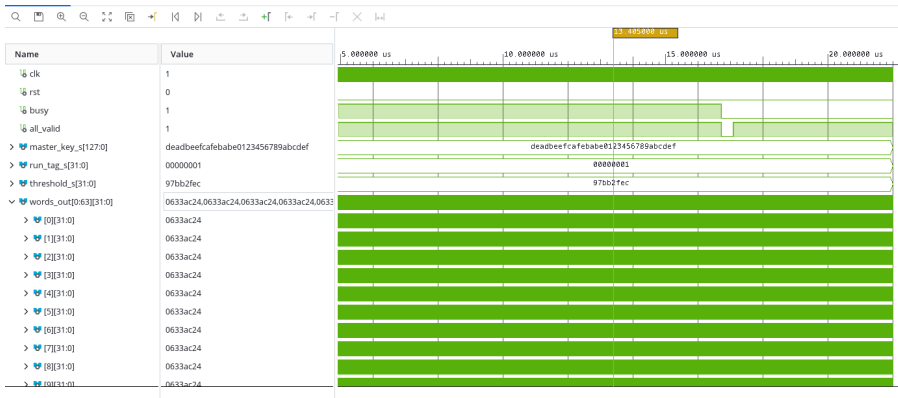
Block Diagram

FPGA System Architecture



RNG Bank — Architecture

- N_{rows} independent Trivium stream ciphers, one per grid column
- N_{rows} is a **synthesis-time generic** (`N_ROWS_G`), default 64
- Synthesized for 64, 128, or 180 — same throughput, different resource usage



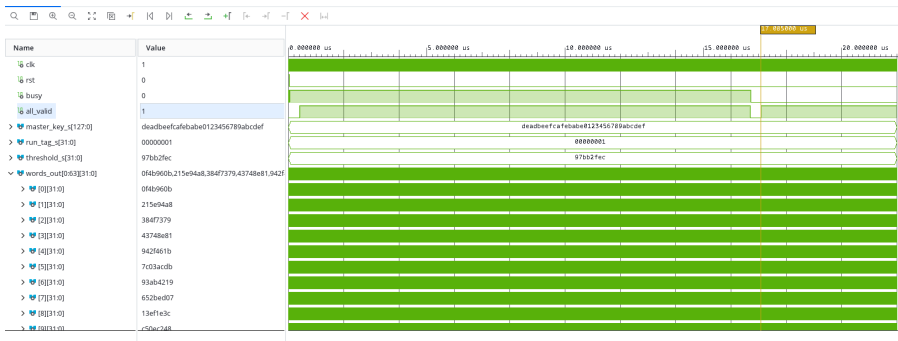
RNG Bank — Seeding & Output

AES-CTR Seeding:

- For $N_{\text{rows}} = 64$: 1536 cycles ($2N$ AES blocks $\times$ 12 cyc)
- Trivium warmup: 36 cycles ($1152 \text{ bits} \div 32$)
- **Total: ≈ 1573 cycles** ($\approx 15.7 \mu\text{s}$ at 100 MHz)

Per cycle output:

- N_{rows} random 32-bit words

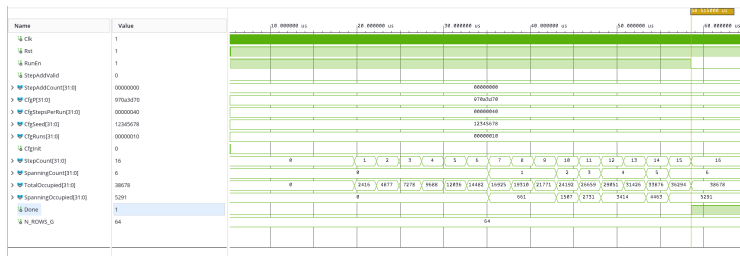


Frontier Engine — Row-Wise Algorithm

Problem: determine if a top-to-bottom path exists without storing the full grid.

Algorithm (per row):

- 1 **Vertical seed:** $\text{seed}[i] = \text{open}[i] \wedge \text{prev_reach}[i]$
- 2 **Horizontal closure:** $\text{reach}[i] = \text{seed}[i] \vee (\text{open}[i] \wedge (\text{reach}[i-1] \vee \text{reach}[i+1]))$

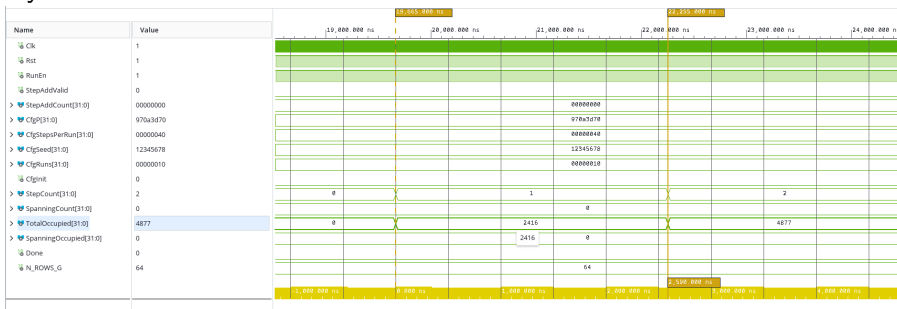


Bidirectional prefix scan: resolves in $\log_2 N_{\text{rows}}$ stages (6 for $N = 64$) in a single cycle

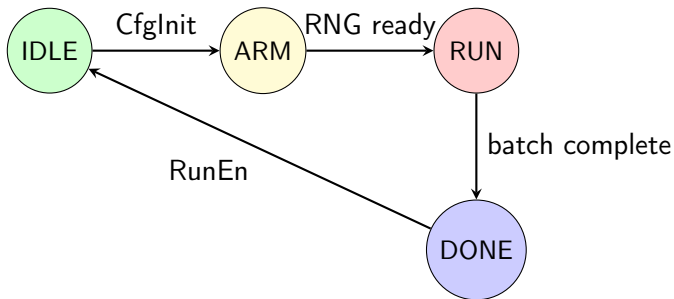
Frontier Engine — VHDL Pipeline

3-stage pipeline:

- RUN_READY → RUN_COMPUTE → RUN_SAVE
- 3 cycles/row core computation
- 4 cycles/row end-to-end (includes registered handshake)
- Only 2 rows of state buffered



Core Controller — State Machine



Batch processing:

- CfgRuns runs executed autonomously — no host intervention
- Accumulators (32-bit): StepCount, SpanningCount, TotalOccupied, SpanningOccupied
- RNG runs continuously — no cycle wasted waiting for random numbers

UART Protocol — 16-Byte Frames

Request (host → FPGA):

Word	Field	Purpose
0	CfgP (UQ32)	Probability p
1	CfgSeed	RNG seed
2	CfgStepsPerRun	Grid height
3	CfgRuns	Batch size

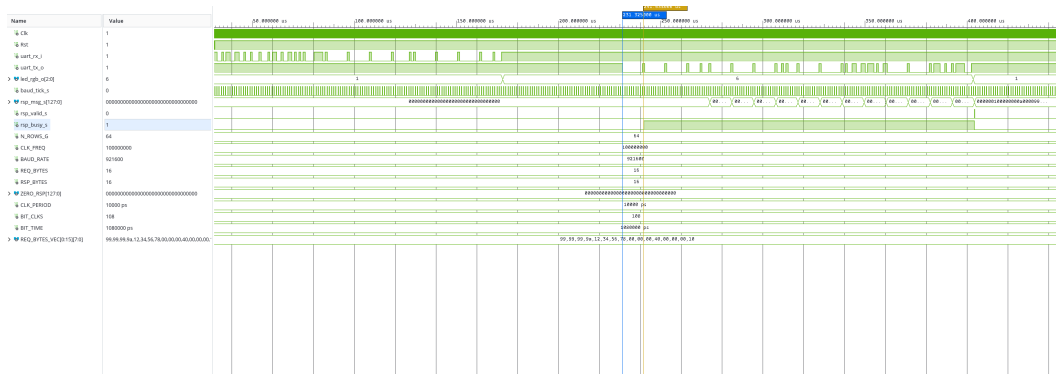
Response (FPGA → host):

Word	Field	Purpose
0	StepCount	Runs completed
1	SpanningCount	Spanning runs
2	TotalOccupied	Total occupied site
3	SpanningOccupied	Reachable sites

Details:

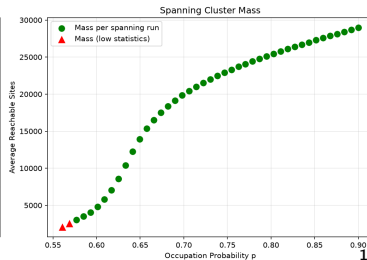
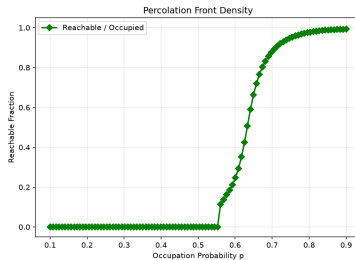
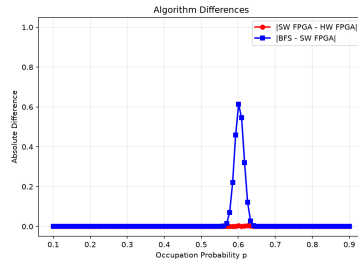
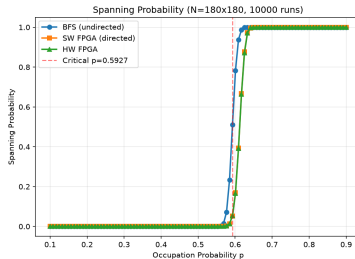
- 115200 baud, 8N1
- Wire time: ≈ 2.78 ms
- Linux termios raw mode
- Fixed-point: $p = \text{CfgP} / 2^{32}$

UART Protocol — End-to-End Simulation



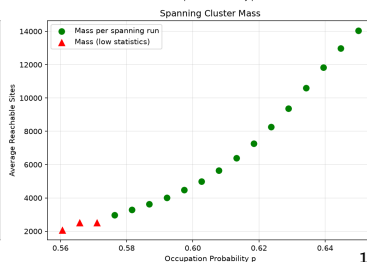
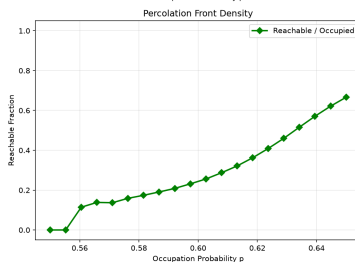
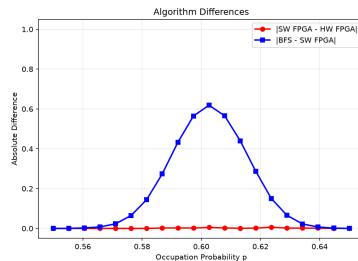
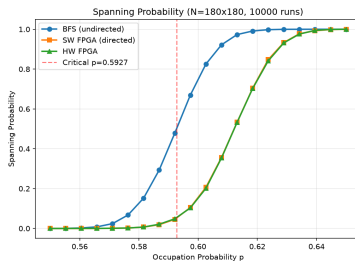
Three-Way Comparison — Full Range

- BFS (undirected) vs SW FPGA (directed) vs HW FPGA
- 20 points, 1000 runs each
- Directed percolation threshold is **higher** ($p_c \approx 0.605$) than isotropic ($p_c \approx 0.593$)

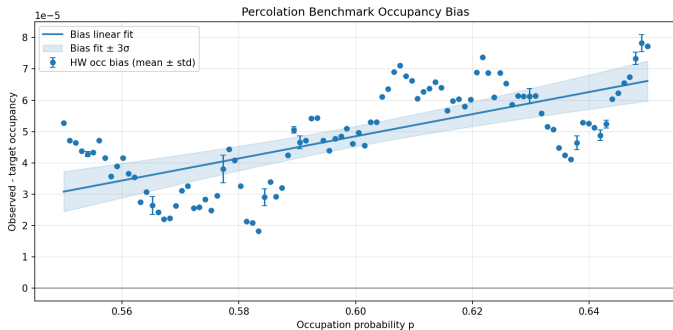


Three-Way Comparison — Threshold Region

- Zoom around the critical region
- 100 points, 1000 runs each
- Perfect agreement between software model and hardware implementation



Occupancy Bias — RNG Accuracy

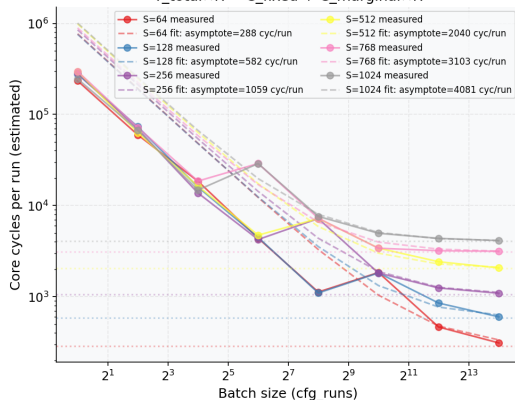


- Bias = $\langle p_{\text{measured}} \rangle - p_{\text{target}}$
- Maximum bias < 0.001 across full p range
- No systematic trend — RNG is unbiased

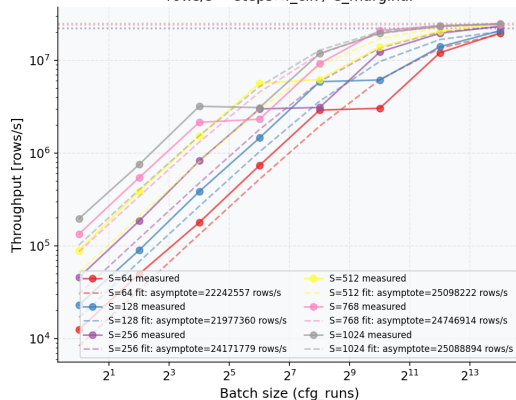
Asymptotic Throughput — Amdahl Decomposition

$N=128, p \approx 0.6$

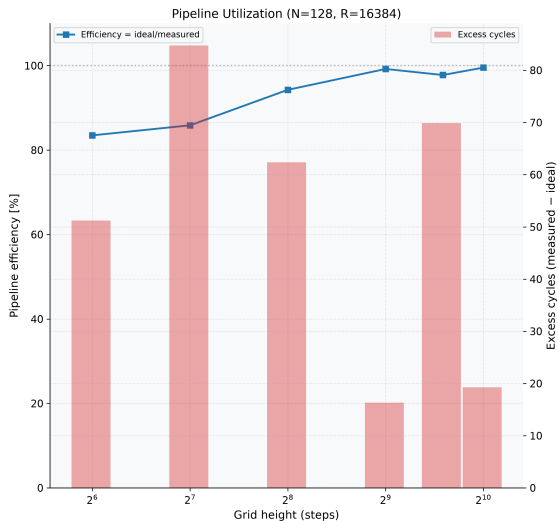
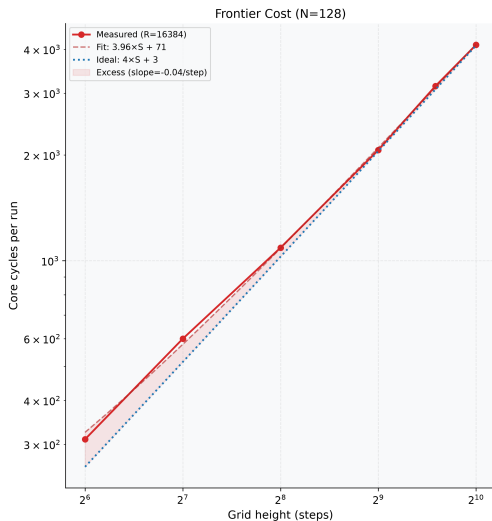
Asymptotic Cycles-per-Run Decomposition
 $T_{\text{total}} \times R = C_{\text{fixed}} + C_{\text{marginal}} \times R$



Asymptotic Throughput (Amdahl)
 $\text{rows/s} \rightarrow \text{steps} \times f_{\text{clk}} / C_{\text{marginal}}$



Theoretical vs practical efficiency



Model and Speed results

Model: $T_{\text{total}} \times R = C_{\text{fixed}} + C_{\text{marginal}} \times R$

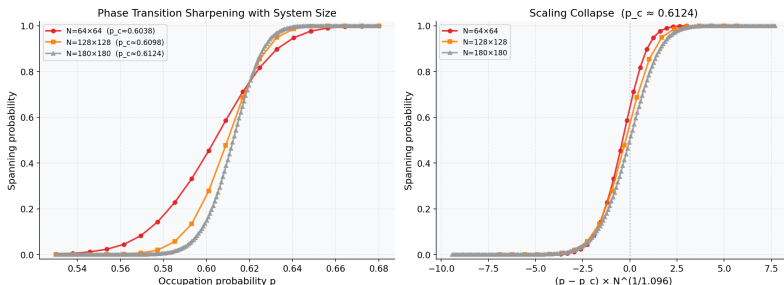
- C_{fixed} : UART + host overhead (one-time per request)
- C_{marginal} : per-run core cost (asymptote at large R)
- **Throughput:** $\frac{\text{rows}}{s} = \frac{\text{steps} \times f_{\text{clk}}}{C_{\text{marginal}}}$

Results:

- Asymptote matches theoretical $4 \times S$ cycles/run (3 frontier + 1 handshake)
- Measured: $C_{\text{core}} = 3.99 \times S + \text{const}$ (vs ideal $4 \times S$)
- For $N = 64$: measured asymptote ≈ 288 cycles/run vs theoretical $4N + 3 = 259$ ($\approx 11\%$ state-machine overhead at the latest point we measured)
- **Efficiency:** $\approx 100\%$ of achievable $4 \times S$ model
- Throughput scales with batch size, saturating at $\sim 2.5 \times 10^7$ rows/s

Finite-Size Scaling — Collapse

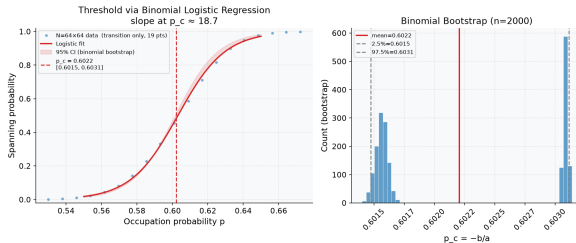
Directed Percolation — Finite-Size Scaling (square grids)



- Square grids: $N = 64, 128, 180$
- Perfect collapse using DP exponent $\nu_{\parallel} = 1.096$
- Confirms the system belongs to the directed percolation universality class

Threshold Estimation — Bootstrap

Directed Percolation Threshold Estimation (64×64, 16k runs/pt)



Method: binomial logistic regression + 2000 bootstrap resamples

$$P_{\text{span}}(p) = \frac{1}{1 + e^{-a(p-p_c)}}$$

- $p_c = 0.6047 \pm 0.0002$ (95% CI)
- Consistent with literature value ≈ 0.605 for DP on 64×64

Thank You

Questions?

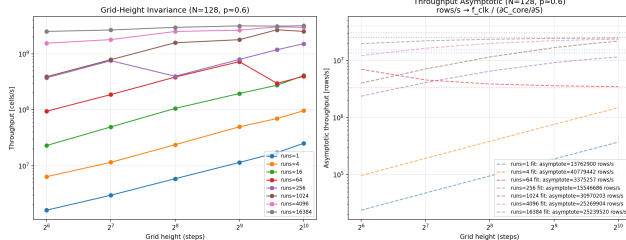
Repository: `https://github.com/.../FPGA-project`

- VHDL: percolation core, RNG bank, frontier engine
- Python: validation, benchmark, analysis
- Full documentation in `project/percolation_core/`

Summary of Results

Metric	Value	Notes
Clock frequency	100 MHz	Single-clock, Artix-7
Grid width N_{rows}	64 (128, 180)	Compile-time generic
Frontier pipeline	3 cycles/row	READY → COMPUTE → SAVE
Cycles per run ($N \times N$)	$4N + 3$	e.g. $N = 64 \rightarrow \approx 259$
Pipeline efficiency	$\approx 100\%$	Of achievable $4 \times S$ model
Throughput	$\sim 10^9$ cells/s	At 100 MHz ($\sim 2.5 \times 10^7$ rows/s $\times 64$)
Occupancy bias	< 0.001	RNG accuracy
Critical threshold p_c	0.6047 ± 0.0002	DP universality class
UART overhead	≈ 2.78 ms	115200 baud, 32 bytes

Grid-Height Invariance & Determinism



Throughput vs grid height

- Throughput **independent** of grid height (pipeline-dominated)
- $CV < 1\%$ across all configurations — FPGA is deterministic
- Tiny jitter is entirely host/USB-subsystem noise

Threshold Estimation — Bootstrap

Directed Percolation Threshold Estimation (180×180, 131k runs/pt)

